

HCI **Postdoctoral Fellow** working on programming-based CAD for personal fabrication, and VR/AR interaction. I design and evaluate interactive creator tools grounded in a strong software/hardware background. I have publications at CHI, UIST, ISMAR, VRST, and SUI (~25% acceptance), including **first-author** work and papers coauthored with mentored MSc/PhD students, and extensive international experience across Europe, Australia, South America, and North America. My work includes research software adopted in real-world tools (e.g., OpenSCAD). I aim to advance interactive systems to improve fabrication experiences.

Education

- 2020 – 2024  **Doctorat (PhD) en Informatique et Applications, Université de Lille (France).**
Supervisors: Thomas Pietrzak and Géry Casiez
Co-tutelle with Carleton University
-  **Doctor of Philosophy (PhD) in Information Technology, Carleton University (Canada).** Supervisor: Audrey Girouard
Thesis: *Facilitating Programming-Based 3D Computer-Aided Design Using Bidirectional Programming.*
- 2016 – 2019  **Master of Science (MSc) in Computer Science, Los Andes University (Colombia).**
Thesis: *Study on improving interactions with objects in virtual reality: Use cases with haptic redirection and active and passive haptic feedback devices.*
◇ **International internship:** Creatis Interaction Lab (CIL), Carleton University (Canada), 2019, 6 months
- 2011 – 2016  **Bachelor's Degree in Systems and Computing Engineering, Los Andes University (Colombia).**
◇ **International internship and exchange:** Centre de Recherche en Acquisition et Traitement de l'Image pour la Santé CREATIS and Institut National des Sciences Appliquées INSA-Lyon (France), 2014 – 2015, 1 year
- 2009 – 2014  **Bachelor's Degree in Electronic Engineering, Los Andes University (Colombia).**

Appointments

- 2026 – present  **Visiting Researcher (3 months), Monash University, Australia.**
Conducting research in Virtual Reality (VR), specializing in haptic feedback and target selection techniques for multi-user, cross-platform collaborative environments.
- 2024 – present  **Postdoctoral Fellow, Carleton University, Canada.**
Leading research on VR/XR interaction, focusing on input devices, avatar representation, and user experience in multi-user, cross-platform collaborative environments.
- 2021 – 2024  **PhD Researcher, Institut National de Recherche en Sciences et Technologies du Numérique (INRIA), France, and Creatis Interaction Lab (CIL), Carleton University, Canada.**
Conducted research in 3D CAD modeling for digital personal fabrication with 3D printers, focusing on OpenSCAD.
◇ Some proposed features from this research were implemented and integrated into the online version of the OpenSCAD application.
- 2019  **Visiting Student Researcher (6 months), CIL, Carleton University, Canada.**
Conducted research on Haptic feedback for Virtual Reality (VR), including hardware and software prototyping to enhance user interaction in virtual environments.

- 2016 – 2018  **Research Assistant**, *Los Andes University, Bogotá, Colombia.*
Designed and developed automation tools to efficiently manage large-scale surveys, streamlining data collection and analysis processes.
- 2014 – 2015  **Research Engineer Intern**, *Centre de Recherche en Acquisition et Traitement de l'Image pour la Santé (CREATIS), France.*
Developed web-based modules enabling the execution of processing algorithms and visualization of medical imaging software directly on the web, facilitating broader accessibility and integration of the BBTK

Teaching Experience

Instructor

Responsible for designing and delivering graduate-level courses, including syllabi, lectures, labs, assessments, and full-class instruction.

- 2025 – 2025  **Carleton University (Canada)**
 ◇ **School of Information Technology (130 hours)**
 ITEC 5207 - Data Interaction Techniques (**Graduate-level data visualization elective**)

Teaching Assistant

Over 1,500+ teaching hours in computer science, mathematics, and electrical engineering, from first-year programming and algorithms to interactive visual computing and data interaction techniques, at Los Andes University and Carleton University. Responsibilities included teaching undergraduate and graduate-level classes, marking, conducting seminars, proctoring examinations, and leading discussions in courses.

- 2021 – 2021  **Carleton University (Canada)**
 ◇ **School of Information Technology (130 hours).**
 OSS 4006 - Image Processing
 OSS 1002 - Optical Systems and Sensors
- 2010 – 2019  **Los Andes University (Colombia).**
 ◇ **Department of Science - Mathematics (+290 hours) :**
 MATE1105 - Linear Algebra (4 times)
 MATE2230 - Numerical Analysis and Complex Variable
 MATE1207 - Vector Calculus
 ◇ **Department of Systems and Computing Engineering (+1150 hours)**
 ISIS1204 - Algorithmics and Object-Oriented Programming I (6 times)
 ISIS1205 - Algorithmics and Object-Oriented Programming II (2 times)
 ISIS2503 - Software architecture and design
 ISIS4823 - Interactive visual computing (**Graduate-level visualization elective**)
 ◇ **Department of Electric and Electronic Engineering (150 hours)**
 IELE1500 - Signals and systems analysis
 IELE2000 - Circuit analysis and synthesis

Employment History

- 2019 – *present*  **IT Contractor**, Los Andes University, Bogotá, Colombia.
Automated data loading, surveying, and result reporting processes for perception studies. Developed C#, JavaScript, Python, Java, and MSSQL tools to streamline operations.

- 2017 – 2019  **Software Engineer**, Openplus Lda, Unitco, Bogotá, Colombia.
Developed and supported IT tools such as Customer Relationship Management (CRM) systems and specialized tools. (Python, MSSQL, VBA).
- 2016 – 2017  **Software Engineer**, Mareigua Ltda, Bogotá, Colombia.
Designed, developed, and tested software solutions in C# and MSSQL.

Publications

Conference Proceedings

- 1 E. DeMarbre, J. Henderson, **J. F. Gonzalez**, and R. J. Teather, “Effects of Virtual Controller Representation and Virtuality on Selection Performance in Extended Reality,” in *31st ACM Symposium on Virtual Reality Software and Technology*, Montreal QC, Canada: ACM, Nov. 2025, ISBN: 979-8-4007-2118-2.  DOI: 10.1145/3756884.3766004, Acceptance Rate: 27%.
- 2 N. Ocampo, **J. F. Gonzalez**, and R. J. Teather, “Comparing Hand and Controller Avatars with Hand Tracking and Controller-Based Interaction,” in *2025 International Symposium on Mixed and Augmented Reality (ISMAR)*, Daejeon, South Korea, Oct. 2025.  DOI: 10.1109/ISMAR67309.2025.00029, Acceptance Rate: 20.6%.
- 3 **J. F. Gonzalez**, T. Pietrzak, A. Girouard, and G. Casiez, “Facilitating the Parametric Definition of Geometric Properties in Programming-Based CAD,” in *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, ser. UIST’24, Pittsburgh, PA, USA: ACM, Oct. 13, 2024, ISBN: 979-8-4007-0628-8/24/10.  DOI: 10.1145/3654777.3676417, Acceptance Rate: 24%.
- 4 **J. F. Gonzalez**, T. Pietrzak, A. Girouard, and G. Casiez, “Understanding the Challenges of OpenSCAD Users for 3D Printing,” in *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)*, Honolulu, Hawaii, USA: ACM, May 11, 2024.  DOI: 10.1145/3613904.3642566, Acceptance Rate: 26%.
- 5 **J. F. Gonzalez**, D. Kieken, T. Pietrzak, A. Girouard, and G. Casiez, “Introducing Bidirectional Programming in Constructive Solid Geometry-Based CAD,” in *Proceedings of the 2023 ACM Symposium on Spatial User Interaction*, ser. SUI’23, Sydney, Australia: ACM, Oct. 13, 2023, pp. 1–12, ISBN: 9798400702815.  DOI: 10.1145/3607822.3614521, Acceptance Rate: 35%.
- 6 **J. F. Gonzalez**, J. C. McClelland, R. J. Teather, P. Figueroa, and A. Girouard, “Adaptic: A Shape Changing Prop with Haptic Retargeting,” in *Proceedings of the 2021 ACM Symposium on Spatial User Interaction*, ser. SUI’21, New York, NY, USA: ACM, Nov. 9, 2021, pp. 1–13, ISBN: 978-1-4503-9091-0.  DOI: 10.1145/3485279.3485293, Acceptance Rate: 39.5%.

Thesis

- 1 **J. F. Gonzalez**, “Facilitating programming-based 3d computer-aided design using bidirectional programming,” Ph.D. dissertation, Université de lille ; Carleton University (CA), Apr. 11, 2024. Accessed: Jul. 29, 2024.  URL: <https://hal.science/tel-04635570>.

Posters

- 1 J. C. McClelland, **J. F. Gonzalez**, R. J. Teather, P. Figueroa, and A. Girouard, “Adaptic: A Shape Changing Prop with Haptic Retargeting,” Poster, presented at the IEEE World Haptics Conference 2019 (Tokio, Japan), Jul. 2019. Accessed: Apr. 15, 2024.  URL: https://www.csit.carleton.ca/~rteather/pdfs/world_haptics2019.pdf.

Awards, Honors & Grants

- 2026  **Mitacs Globalink Research Award - Canada.** Awarded CAD \$6,000 in travel support for a research visit at *Monash University*, Melbourne, Australia.
- 2025  **Professional Development Fund — Carleton University Postdoctoral Union.** Awarded CAD \$1,300 in travel support for an academic visit and invited talk at *Universidad de los Andes*, Bogotá, Colombia.
-  **Prix de thèse Interaction Humain–Machine** (*Association Francophone d'Interaction Humain–Machine, AFIHM*) — *Nomination*. Supervisor-nominated for the HCI doctoral thesis prize (France).
-  **Bill Buxton Award** (*Graphics Interface Conference*) — *Nomination*. Supervisor-nominated for outstanding doctoral dissertation in HCI (Canada).
- 2024  **Carleton University Senate Medal for Outstanding Academic Achievement (Doctoral)** — *Nomination*. Nominated by the PhD defense jury committee.
- 2020  **Région Hauts-de-France — Doctoral Grants Competition.** Co-tutelle PhD funding supporting research at *Université de Lille* (France) in collaboration with Carleton University.
-  **NSERC Alliance & Ontario Centre of Excellence (OCE).** Contributed as a graduate researcher to funded projects supported by NSERC Alliance and OCE (Canada).
- 2019  **Emerging Leaders in the Americas Program (ELAP).** Awarded CAD \$9,000 for a 6-month **Visiting Student Researcher** position at Carleton University, Canada.
 ◇ Selected for the *ELAP Academic Tour* (Toronto, Ottawa, Québec City).

Mentoring and Supervision

- 2025 (6 months)  **Mentor in the program "Quiero Mentoría",** Universidad de los Andes, Colombia. Career/vocational mentoring for last-semester undergraduates.
- 2024 – present  **Co-supervision of PhD and Master's researchers,** Carleton University. Co-mentor (with faculty supervisors) two PhD and one Master's students on extended reality (XR) interactions covering study design, ethics, analysis, and writing.
 ◇ *Outcomes:* two peer-reviewed papers (co-author) [1], [2] , one submitted manuscript.

Academic Service

- 2025  **Reviewer – Full Paper**
ACM Conference on Human Factors in Computing Systems (CHI'25).
 ◇ *Special Recognition for Outstanding Review*
ACM Symposium on Virtual Reality Software and Technology (VRST'25).
ACM Symposium on Spatial User Interaction (SUI'25).
ACM Symposium on User Interface Software and Technology (UIST'25).
 ◇ *Special Recognition for Outstanding Review*
-  **Area Chair (1AC - 6 submissions) – Late-Breaking Work** ACM International Conference on Mobile Human-Computer Interaction (MobileHCI'25).
- 2022  **Reviewer – Work-in-Progress** ACM International Conference on Tangible, Embedded, and Embodied Interaction (TEI'22).

Invited Talks

- 2025  **HCI Seminar**, Universidad de los Andes, Bogotá, Colombia. Academic visit and seminar on programming-based CAD and XR interaction.
- 2021  **Panelist, International Research Mobility Symposium**, Carleton International — Carleton University, Ottawa, Canada.
- 2019  **Panelist, Conference of the Americas on International Education (CAEI)**, Corferias, Bogotá, Colombia.
-  **Panelist, ExpoSolar Colombia**, Plaza Mayor, Medellín, Colombia.

Skills

- Languages  English (C1 - CELPIP), French (B2/C1 - TCF), Portuguese (Intermediate), Spanish (Native).
- Coding  Java, C#, C++, C, Python, SQL, Javascript, VBA, VHDL, $\LaTeX$, MatLab, MevisLab, Unity, Arduino, git.
- Databases  MySQL, MSSQL, SQLITE
- Misc.  Unity, OrCAD, Academic research, mathematical and programming skills, image processing, teaching, $\LaTeX$ typesetting and publishing.